

MCA Python Assessment

Instructions:

- All questions are compulsory.
- Mention proper comments in the code.
- Use correct and meaningful identifiers.
- Mention the expected output at the end of each program.
- Submit all solutions in a GitHub repository named 'Python_assessment'.

Q1. Write a Python program that accepts a paragraph from the user and calculates the number of uppercase letters, lowercase letters, digits, spaces, and special characters. Display the result in descending order based on frequency.

Q2. Develop a grading system where marks of five subjects are entered by the user. The program should calculate percentage, assign grades using nested if-else conditions, and identify whether the student qualifies for a scholarship.

Q3. Create a program that generates the first N prime numbers using a for loop and also calculates the sum and average of those prime numbers.

Q4. Write a menu-driven banking application using a while loop that allows users to deposit, withdraw, check balance, and exit only when the user chooses the exit option.

Q5. Design a Python function that accepts a list of integers and returns the second largest and second smallest element without using built-in sorting functions.

Q6. Write a Python program to merge two lists, remove duplicate values, sort the final list in descending order, and display only those numbers that are divisible by both 3 and 5.

Q7. Create a program that stores employee records in tuples. Each tuple should contain employee ID, name, and salary. Display employees whose salary is above the average salary.

Q8. Write a Python program to perform union, intersection, symmetric difference, and subset operations on two sets entered by the user.

Q9. Develop a dictionary-based inventory management system where users can add products, update quantity, search products, and display low-stock items.

Q10. Write a program to check whether a given string is a palindrome after removing spaces, punctuation marks, and converting all characters to lowercase.

Q11. Create a text file containing student marks. Read the file and display the topper, average marks, and students scoring below average.

Q12. Develop a calculator program that handles division by zero, invalid inputs, and incorrect operations using multiple exception blocks.

Q13. Write a NumPy program to create a 5x5 matrix with random integers and calculate row-wise sum, column-wise sum, transpose, and determinant.

- Q14.** Create a line chart using Matplotlib to display monthly sales data of a company. Add title, labels, legend, and grid for better visualization.
- Q15.** Write a Pandas program to read employee data from a CSV file and display department-wise average salary and highest salary employee.
- Q16.** Write a Python program using functions and loops to generate Fibonacci numbers up to N and store only even Fibonacci numbers in a list.
- Q17.** Create a program that checks whether numbers entered by the user are Armstrong numbers using loops and conditional statements.
- Q18.** Develop a frequency counter that stores occurrence of each word from a paragraph into a dictionary and displays the most repeated word.
- Q19.** Write a function that accepts a list of integers and returns a new list containing factorial of only even numbers.
- Q20.** Create a file processing program that handles file not found errors and counts total words, lines, and characters in a text file.
- Q21.** Generate a NumPy array of student marks and convert it into a Pandas DataFrame. Display highest marks, average marks, and subject-wise statistics.
- Q22.** Read sales data from a CSV file using Pandas and plot a bar chart representing product-wise sales using Matplotlib.
- Q23.** Write a program to print a pyramid pattern of numbers and calculate the sum of all numbers printed in the pattern.
- Q24.** Create a function for division of two numbers and use exception handling to validate inputs and avoid runtime errors.
- Q25.** Store word frequencies from a text file into a dictionary and display the top five most frequently used words.
- Q26.** Write a program to separate vowels and consonants from a sentence and store them in different lists.
- Q27.** Create tuples containing student names and courses. Convert them into sets to identify students enrolled in multiple courses.
- Q28.** Develop a menu-driven mathematical utility program using functions for factorial, prime checking, and Armstrong number checking.
- Q29.** Create a DataFrame containing student marks in multiple subjects. Calculate total marks, percentage, and assign grades using `apply()` function.
- Q30.** Write a NumPy program to create two matrices and perform addition, subtraction, multiplication, and inverse operations.
- Q31.** Plot a pie chart showing market share of different smartphone brands and highlight the brand with maximum market share.
- Q32.** Write a Python program that validates user age for voting eligibility and handles invalid numeric and non-numeric inputs gracefully.

Q33. Create a program that copies contents from one file to another and displays the total number of words copied.

Q34. Write a program to encrypt a message using Caesar Cipher technique and decrypt it back to the original message.

Q35. Develop a contact management system using list and dictionary where users can add, update, search, and delete contacts.

Q36. Create an ATM simulation program that repeatedly asks for PIN validation and locks the account after three invalid attempts.

Q37. Write a recursive function to calculate the sum of digits of a given number and compare it with iterative approach.

Q38. Generate random temperature data using NumPy and plot a histogram representing temperature distribution.

Q39. Read a CSV file containing student attendance records and display students with attendance below 75%.

Q40. Design a mini student management system using functions, dictionaries, file handling, exception handling, and Pandas for report generation.